Machine Learning

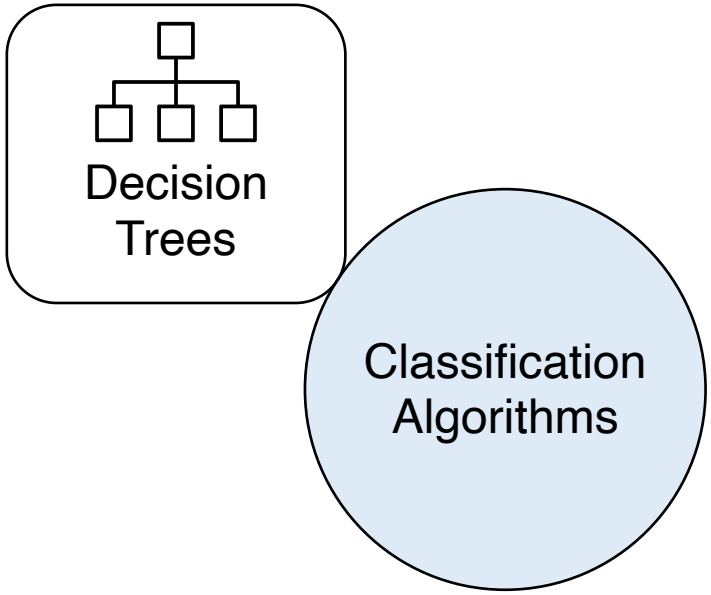
Lecture 4: Bayesian Classification

Spring 2025

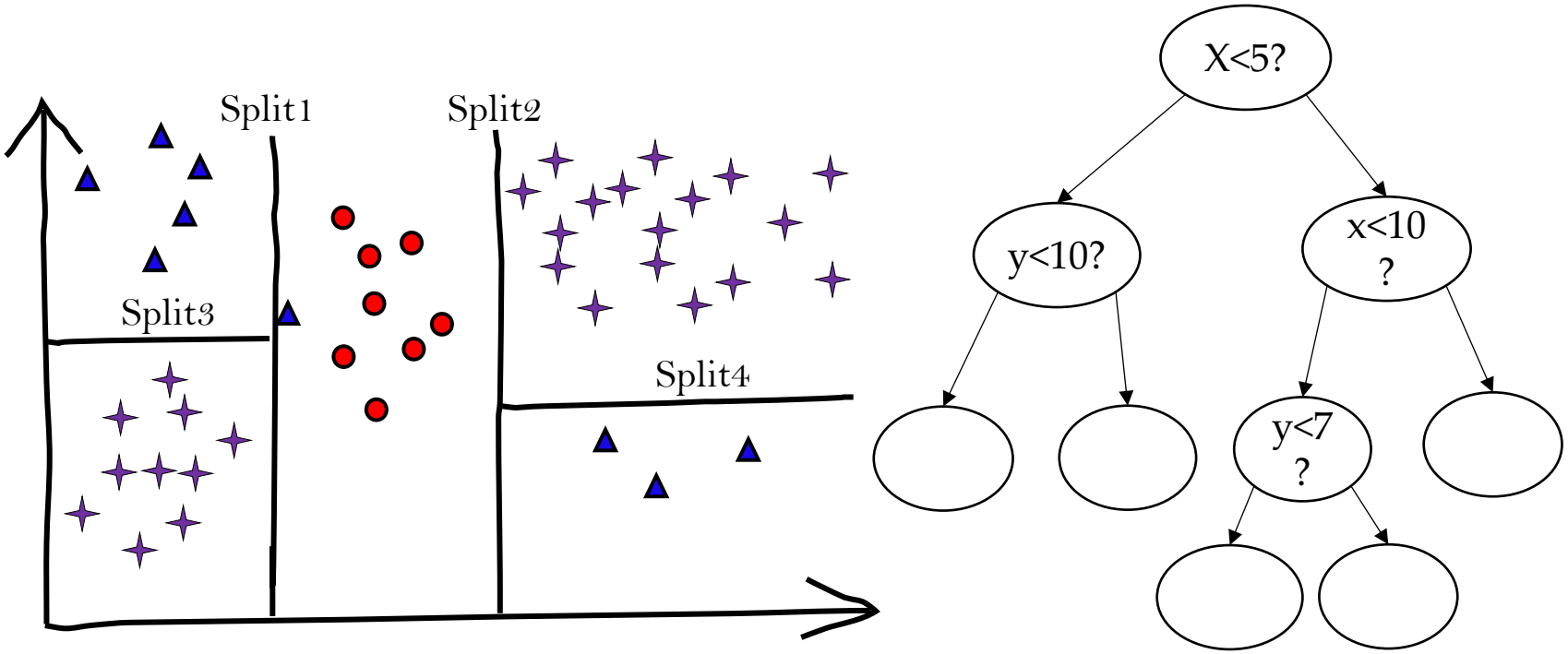
Instructor: Xiaodong Gu



The family of classification

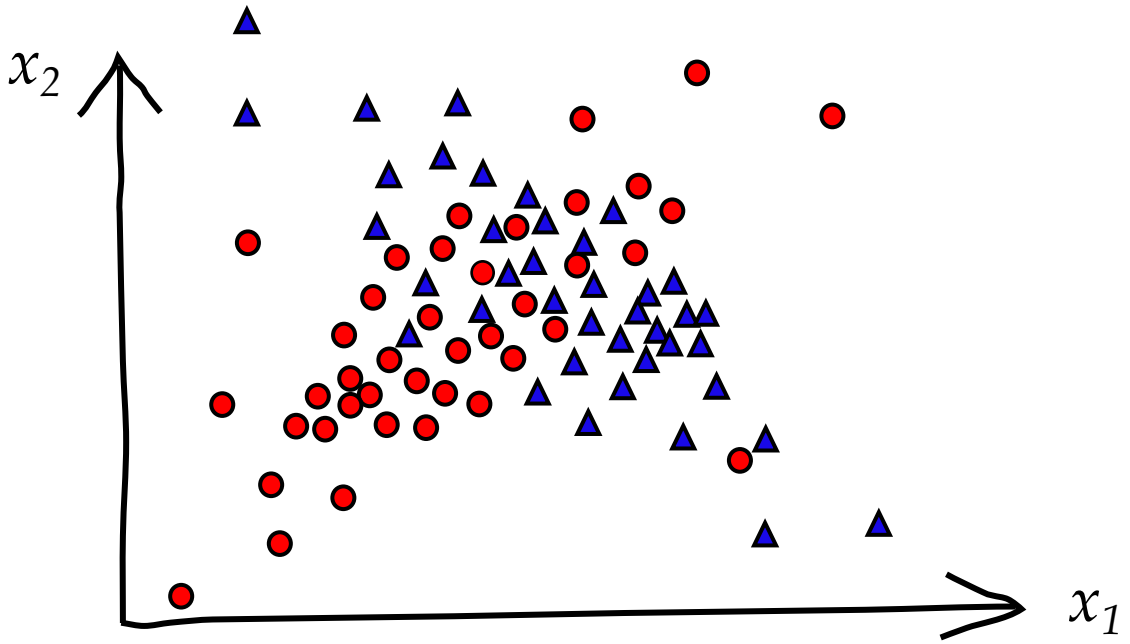


Review: Classification by Decision Trees



Other perspectives on the data?

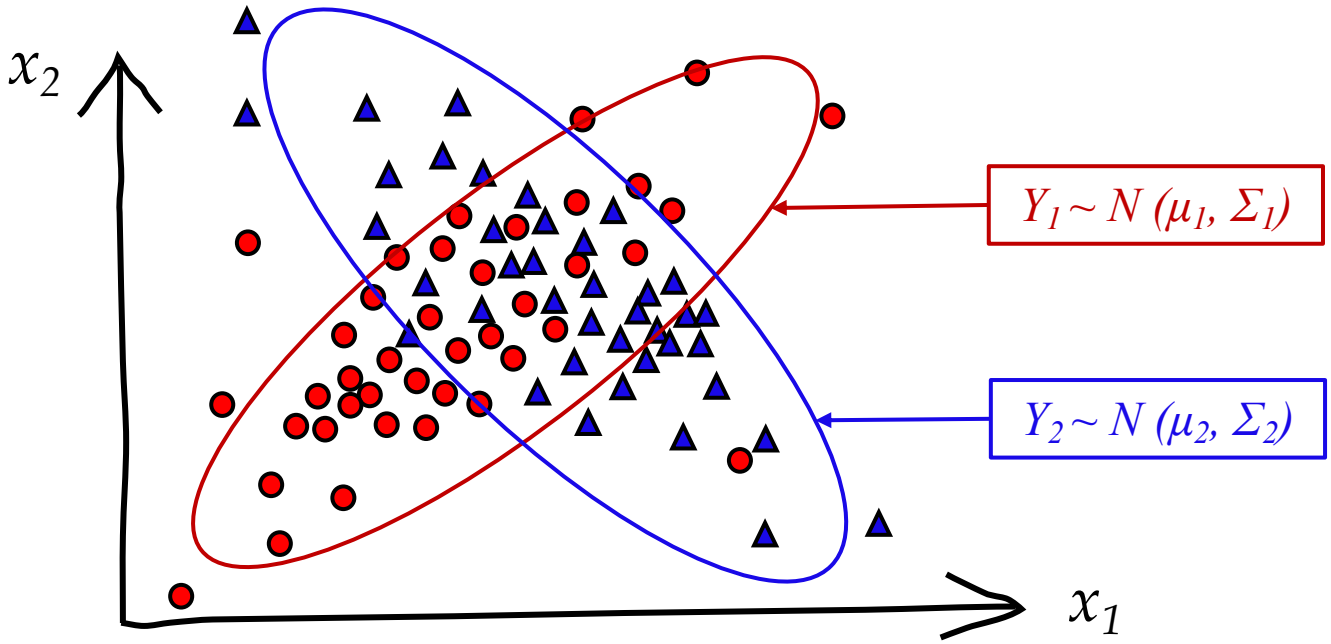
What if the data attributes look like this?



- Can not be separated linearly or recursively)

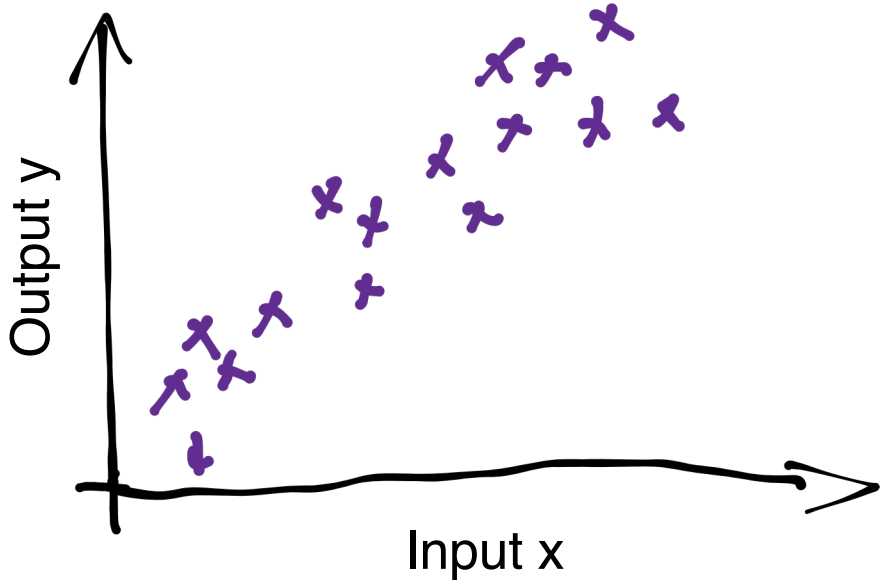
Other perspectives on the data?

What if the data attributes look like this?

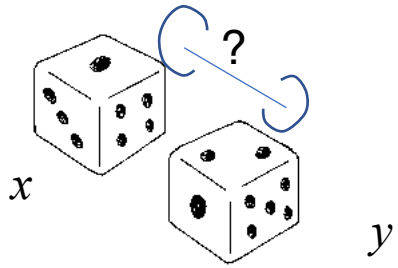


- Can not be separated linearly or recursively)
- But have a clear pattern of probabilistic distribution and dependence

A Probabilistic View of Machine Learning



- model? $p(x, y)$



- parameters?

x	y	$\mathbf{Pr}(x, y)$
1	1	0.4
1	2	0.1
…	…	…

- loss function?
- optimization algorithm?

Machine learning can also be viewed as inferring the **probabilistic relationship** between data features.

Recall: Probabilistic Inference

Data:

Random variables

Major	Age	Browsing
CS	21	Taobao
Physic	22	Github
Math	21	Zhihu
Med	23	Github

Probability a student likes GitHub given that he/she is major in CS?

Joint Prob:

Major	Age	Browsing	Prob
CS	21	Taobao	0.2
Physic	22	Github	0.2
Math	21	Zhihu	0.15
Med	23	Github	0.32

$$P(\text{Browsing} = \text{GitHub} \mid \text{Major} = CS)$$
$$= \frac{P(\text{Browsing} = \text{GitHub} \,\&\, \text{Major} = CS)}{P(CS)}$$
$$= 0.2$$

Bayes rule

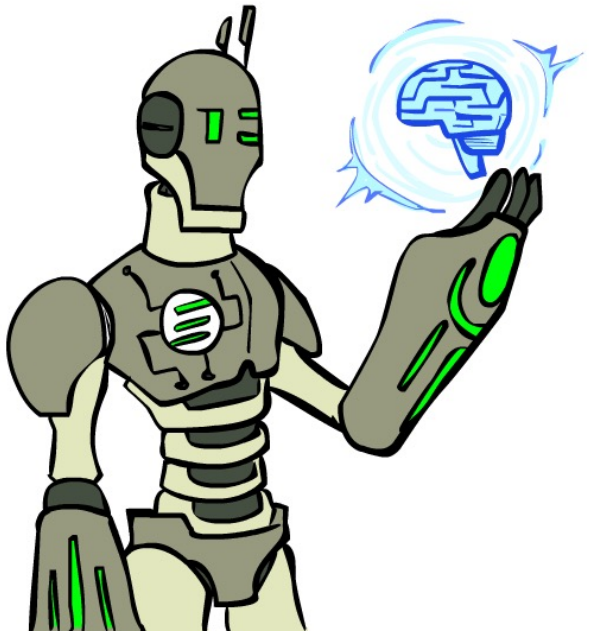
Joint distribution is sufficient to answer **any** probabilistic inference questions of involved variables.

Then, how about classification?

Today

A Probabilistic View of Classification

- Bayesian Decision
- Naïve Bayes Classifier



A simple probabilistic decision

Example: Spam Filtering

We have an inbox of 100 emails with **85** normal and **15** spam messages. If a new e-mail is randomly picked from this inbox, which group (normal or spam) will you guess it is from?

- 2 classes (categories):
  - C_1 = normal; C_2 = spam
- Prior probabilities: $P(C_1)$ and $P(C_2)$

$$P(C_1) = \frac{85}{85+15} = 85\% \qquad P(C_2) = \frac{15}{85+15} = 15\%$$



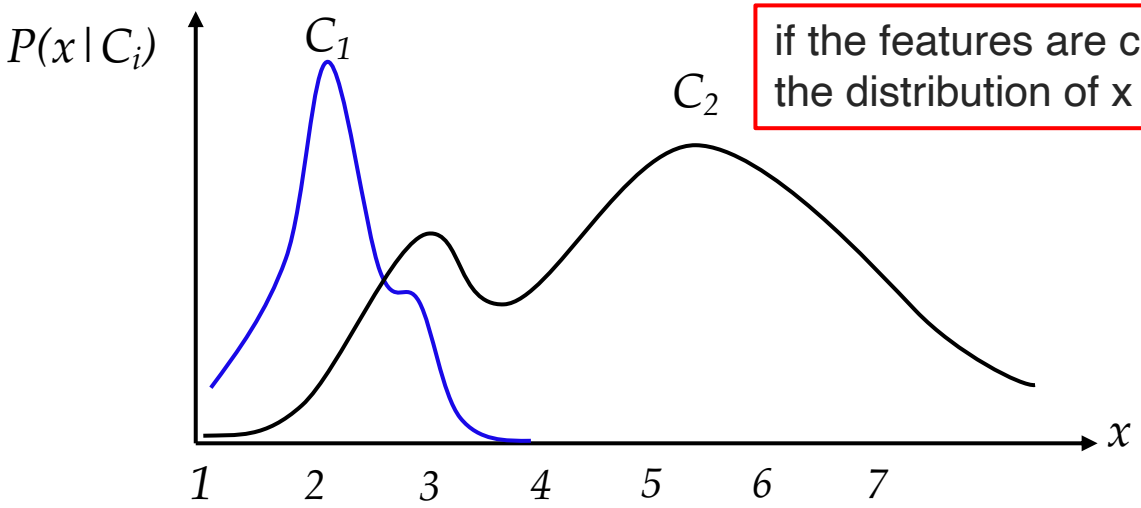
- Decision rule: decide C_1 if $P(C_1) > P(C_2)$
  - always predict that the email comes from normal
  - no need to check the email.

What if we have more information?

Suppose we have an checked their numbers of money-related words…
(e.g., $, 100, million, discount, invoice, investment, etc.).



- data feature $\boldsymbol{x}$ (# money-related words)
- $p(x \mid C_i)$: likelihood (class conditional probability distribution).



if the features are chosen appropriately, the distribution of x should **depend on** C.

Applying Bayes Rule

The likelihood (class-conditional density) of x in each class C_i

The prior probability of class C_i (before observing x)

- Can be simply estimated by frequencies on the training set

$$P(C_i|x) = \frac{p(x|C_i)P(C_i)}{p(x)}, \quad i=1, 2$$

the posterior probability of class C_i (after observing x)

The probability (evidence) that data x will be observed

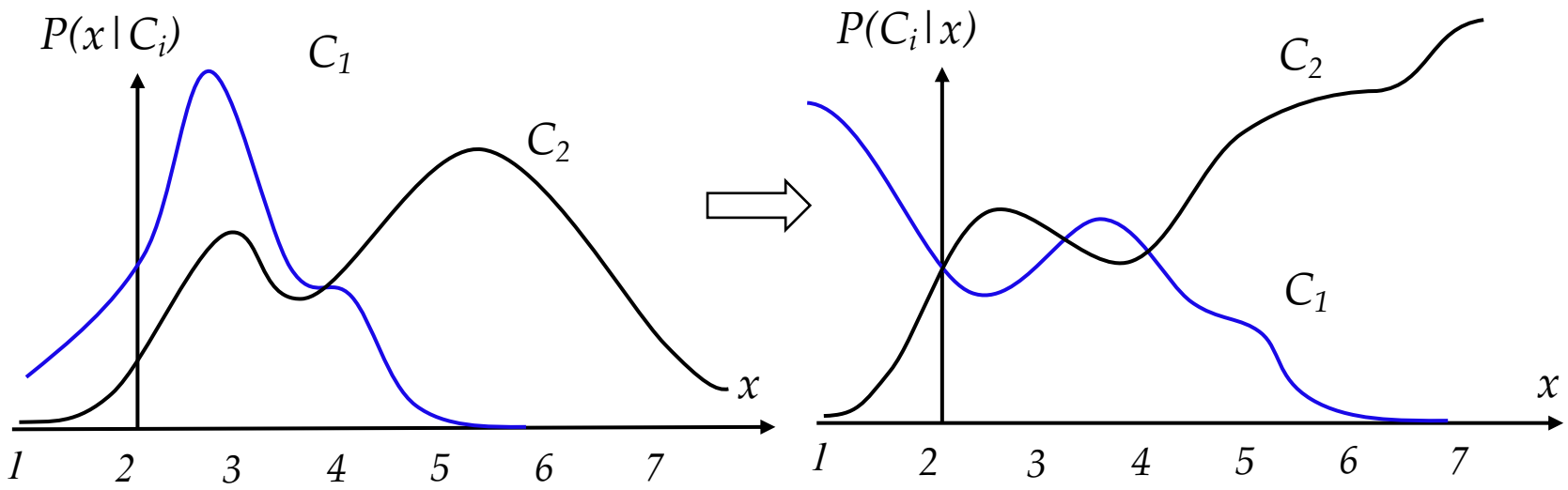
$p(x) = \mathrm{P}(C_1)\mathrm{p}(x \mid C_1) + \mathrm{P}(C_2)\mathrm{p}(x \mid C_2)$

- By MAP, $\widehat{C} = \arg\max_i p(C_i|x)$

Decision rule: decide C_1 if $p(x|C_1)P(C_1) > p(x|C_2)P(C_2)$

Example

- $P(C_1) = 2/3$, $P(C_2) = 1/3$



Example

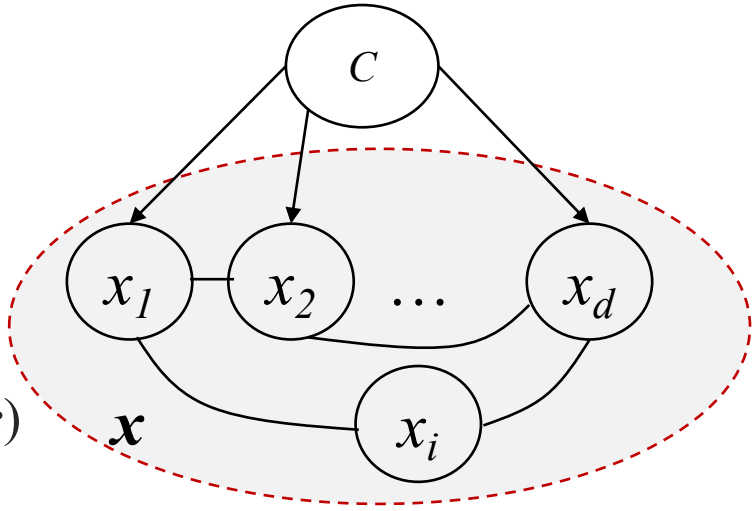
- At $x = 6$, $p(C_1 | x) = 0.92$ and $p(C_2 | x) = 0.08$
- Intuitively, we inclined to decide that the correct class is C_1

What if x has multiple dimensions?

X_1: address	X_2: topic	X_3: length	X_4:	C: class
Irregular	CS	<25	…	spam
Regular	Finance	>25	…	normal
…	…	…	…	…
Irregular	CS	<25	…	spam

- The most probable class $c \in C$:

$$c_{\text{MAP}} = \arg\max_{c \in C} P(c|x)$$
$$= \arg\max_{c \in C} \frac{P(x|c)P(c)}{P(x)}$$
$$= \arg\max_{c \in C} P(x|c)P(c)$$
$$= \arg\max_{c \in C} p(x_1, x_2, \ldots, x_{|d|}|c)\, P(c)$$



Bayesian Networks for Classification

$$c_{\text{MAP}} = \arg\max_{c \in C} p(x_1, x_2, \ldots, x_d | c) P(c)$$

Question

How to estimate $P(c)$ and $P(x_1, x_2, \ldots, x_d | c)$?

- $\hat{P}(c) \leftarrow \dfrac{\text{count}(\text{C=c})}{N}$

n_c: # of training samples for which C=c
N: total # of training samples

- $\hat{P}(x_1, x_2, \ldots, x_d | c) \leftarrow$



too complicated

x_1	x_2	…	x_d	p
…	…	…	…	…
…	…	…	…	…
…	…	…	…	…
…	…	…	…	…

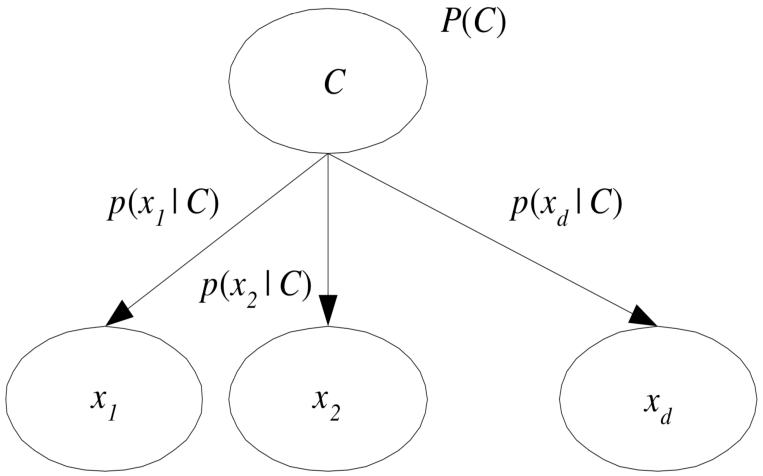
Too many parameters in the joint probability table

Naïve Bayes Independent Assumption

Conditional Independence: assume the input features x_j are **independent** given the class c

$$P(x_1,\ldots,x_d\,|c) = P(x_1|c)\boldsymbol{P(x_2|x_1,c)\ldots P(x_d|x_1,\ldots x_{d-1},c)}$$

$$\approx P(x_1|c)\bullet P(x_2|c)\bullet P(x_3|c)\bullet \ \ldots\ \bullet P(x_d\,|c)$$



$$c_{\mathrm{MAP}} = \arg\max_{c\in C} p(x_1, x_2, \ldots, x_d|c)P(c)$$

$$c_{\mathrm{NB}} = \arg\max_{c\in C} P(c)\prod_{i=1}^{d} p(x_i|c)$$

Training

$$c_{\text{NB}} = \arg\max_{c \in C} P(c) \prod_{i=1}^{d} p(x_i|c)$$

- Training amounts to estimating parameters: $P(c)$'s, $\text{P}(x_1|c), \ldots, \text{P}(x_d|c)$ from data.

Q: How to estimate each P(c)?

– Straightforward

Q: How to estimate $\text{P}(x_i|c)$ for each c?

$$\widehat{P}(x_i|c) \leftarrow \frac{\text{count}(x_i, c)}{\sum_{x \in |x|} \text{count}(x, c)}$$

training samples for which C=c and x = x_i

training samples for which C=c

Zero Counts

Question

What if none of the training instances with class c have attribute x_i?

$$\hat{P}(x_i|c) = 0 \quad \rightarrow \quad \hat{P}(c)\prod_i \hat{P}(x_i|c) = 0$$

no chance to be classified as c, even if all other attributes values suggest c

- **Laplace Smoothing**: add a **virtual count** of 1 to each attribute value.

$$\hat{P}(x_i|c) \leftarrow \frac{\text{count}(x_i, c) + 1}{\sum_{x \in |x|} \text{count}(x, c) + 1}$$

|x| = Vocabulary (the number of different values of attribute x).

The Naïve Bayes Algorithm

Naive_Bayes_Learn(*examples*)

begin
  for *each class* c **do**
    $\hat{p}(c) \leftarrow$ estimate $p(c)$
    for *each attribute value* x_i *of each attribute* x **do**
      $\hat{p}(x_i|c) \leftarrow$ estimate $p(x_i|c)$;
    end
  end
end

Classify_New_Instance(x)

begin

$$c_{\mathrm{NB}} = \arg\max_{c \in C} P(c) \prod_{i=1}^{d} p(x_i|c)$$

end

Applications

- **Text Classification** (e.g., spam filtering, sentiment analysis):

  NB is mostly used in text classification (due to better result in multi-class problems and independence rule) have higher success rate as compared to other algorithms.

- **Real-Time Prediction:**

  NB is an eager learning classifier and it is super **fast**.

- **Multi-Class Prediction:**

  NB can predict the probability of multiple classes of the target variable.

Naïve Bayes for Text Classification



Input:

- a training set of *m* hand-labeled documents $(d_1,c_1),....,(d_m,c_m)$
  where $d_i = \{w_1, w_2,\ldots,w_{|d|}\}$, and $c_i \in C = \{c_1,\ldots,c_{|C|}\}$

Training:

- from training corpus, extract *Vocab*
- calculate $P(c_j)$ terms

  for each c_j in C **do**

  $docs_j \leftarrow$ all docs with class $=c_j$

  $$P(c_j) \leftarrow \frac{|\text{docs}_j|}{|\text{total \# documents}|}$$

- calculate $P(w_k \mid c_j)$ terms

  $text_j \leftarrow$ single doc containing all $docs_j$

  for each word w_k in *Vocab* **do**

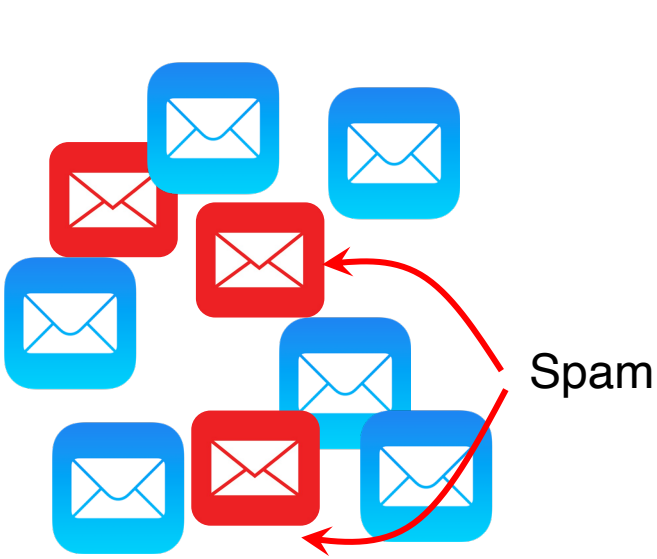
  $n_k \leftarrow$ # of occurrences of w_k in $text_j$

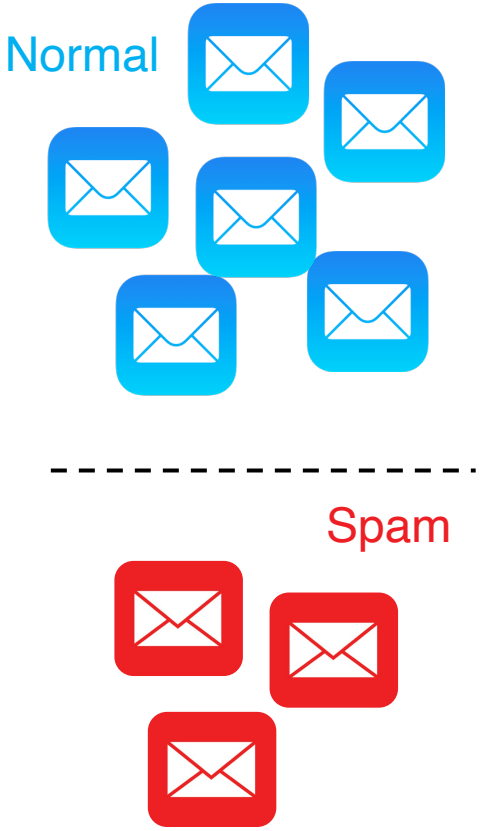
  $$P(w_k \mid c_j) \leftarrow \frac{n_k + \alpha}{n + \alpha\,|Vocab|}$$

Test:

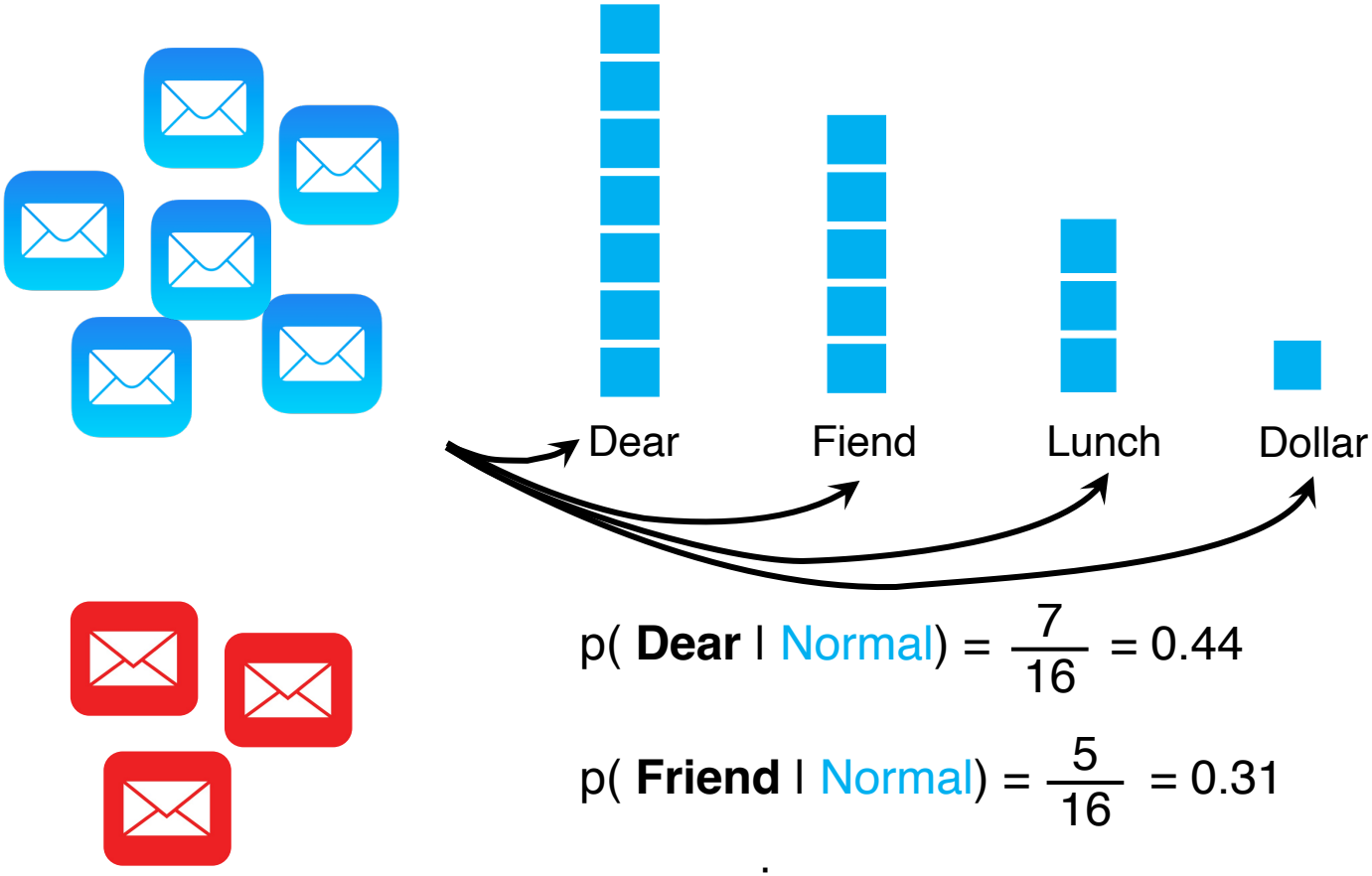
- for $d = \{w_1, w_2,\ldots,w_{|d|}\}$
- For each $c \in C$, calculate $\text{score}(c) = p(c)p(w_1|c)p(w_2|c),\ldots,p(w_{|d|}|c)$
- Output c with the maximum score.

Example: Spam Filtering





Example: Spam Filtering



$$p(\textbf{Dear} \mid \text{Normal}) = \frac{7}{16} = 0.44$$

$$p(\textbf{Friend} \mid \text{Normal}) = \frac{5}{16} = 0.31$$

$$\vdots$$

Example: Spam Filtering

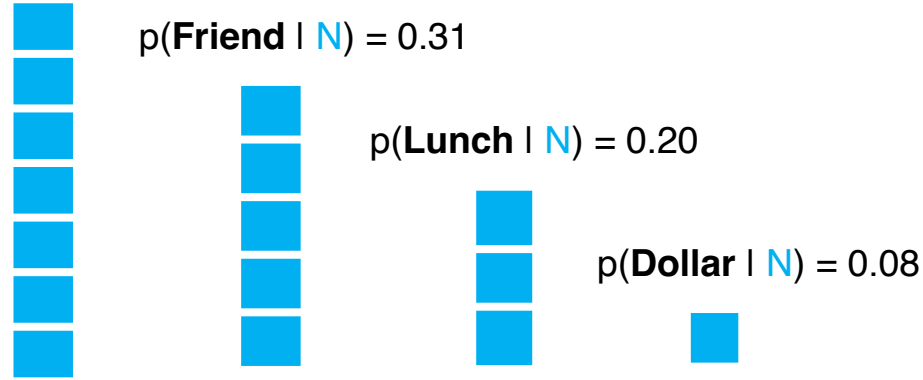


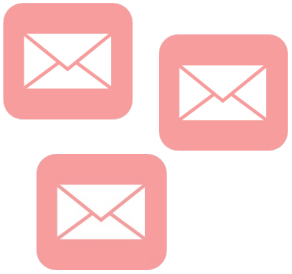
p(**Dear** | N) = 0.44

p(**Friend** | N) = 0.31

p(**Lunch** | N) = 0.20

p(**Dollar** | N) = 0.08





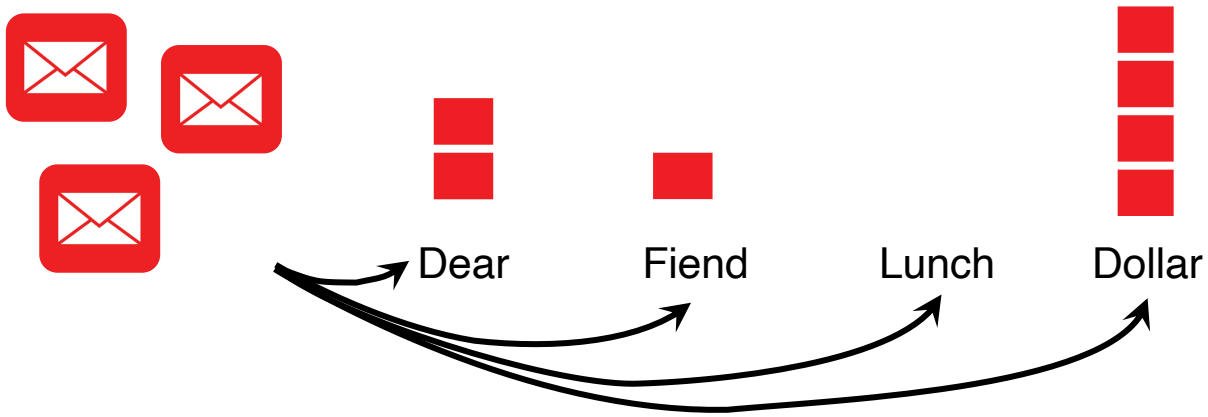
Example: Spam Filtering



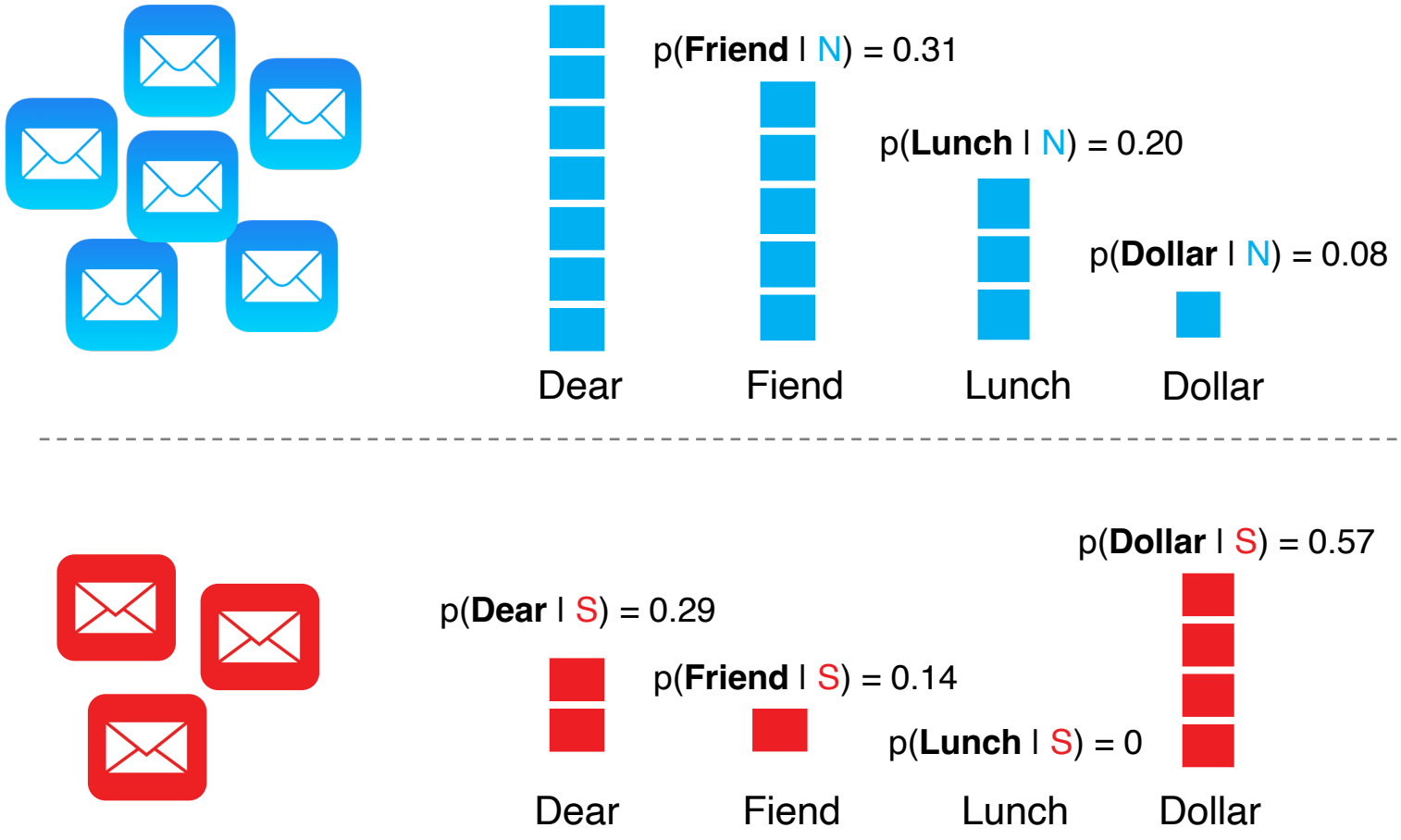
$$p(\textbf{Dear} \mid \text{Spam}) = \frac{2}{7} = 0.29$$

$$p(\textbf{Friend} \mid \text{Spam}) = \frac{1}{7} = 0.14$$

$$p(\textbf{Lunch} \mid \text{Spam}) = \frac{0}{7} = 0$$



Example: Spam Filtering



Example: Spam Filtering

Prior



$$p(N) = \frac{6}{6+3} = 0.67$$



$$p(S) = \frac{3}{6+3} = 0.33$$

Likelihoods

p(**Dear** | N) = 0.44

p(**Friend** | N) = 0.31

p(**Lunch** | N) = 0.20

p(**Dollar** | N) = 0.08

p(**Dear** | S) = 0.29

p(**Friend** | S) = 0.14

p(**Lunch** | S) = 0

p(**Dollar** | S) = 0.57

Posterior

a new message:

"dear friend"

Normal?

Spam?

⇩

P(N | **Dear**, **Friend**)
∝ P(N) × P(Dear | N) × P(Friend | N)
= 0.67 × 0.44 × 0.31
= 0.09

P(S | **Dear**, **Friend**)
∝ P(S) × P(Dear | S) × P(Friend | S)
= 0.33 × 0.29 × 0.14
= 0.03

⇩

Normal

Advantages



- **Fast**
  - ▻ on training, requires only a single pass over the training set
  - ▻ on testing, also fast
- **Competitive performance**
  - ▻ when assumption of independence holds, NB performs better
  - ▻ it also perform well in multi class prediction
- **Simple to update upon additions or deletions of training examples**
  - ▻ easy to maintain

Disadvantages

Conditional Independence Assumption

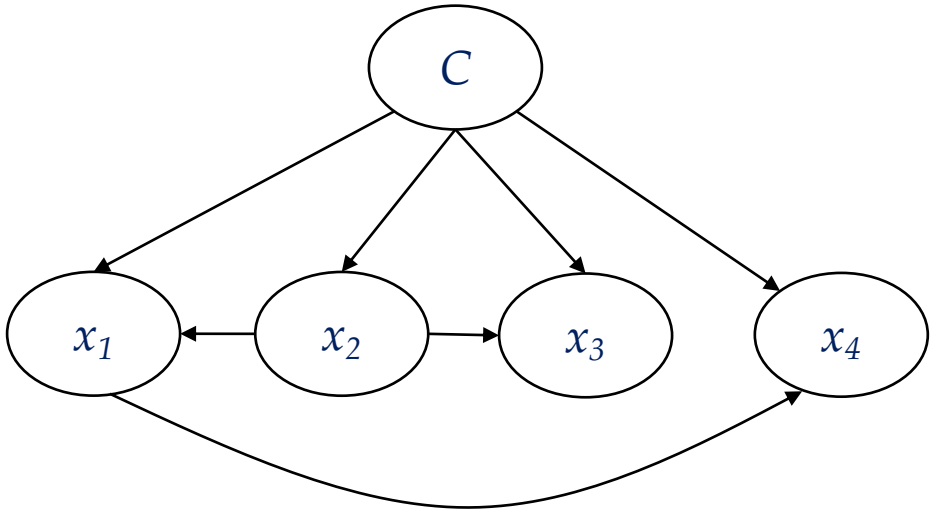
- Often violated
- But it works surprisingly well anyway!
- Don't need estimated posterior $\hat{P}(c|x)$ to be correct
- Need only that

$$\arg\max_{c \in C} \hat{P}(c) \prod_i \hat{P}(x_i|c) = \arg\max_{c \in C} P(c) P(x_1, \ldots, x_d|c)$$

Disadvantages

Underfitting

- The complexity of Naïve Bayes classifier is fixed and low.
- Bayesian (belief) network classifier can relax the assumption.



TIME for Coding



Tutorial : Naïve bayes classifier from scratch

Prompt DeepSeek

“你正在讲授一门机器学习课程，请用原生python写一个朴素贝叶斯模型的tutorial，使用垃圾邮件分类场景”

What's Next?

The Nearest Neighbor Classifier

Classify data by a plurality vote of its nearest neighbors.

- No model (hypothesis) at all !
- Simply memorizing raw data



